Lecture 11

Poisson manifolds and the rigid body

Objectives. *Lecture 10 divided symmetries out of canonical systems. Dividing the rotations out of the rigid body leaves three variables, the components of its body angular momentum. Their bracket has a matrix $\Pi(x)$ that varies with the point and is singular everywhere. Its phase spaces are the spheres centred at the origin, together with the origin itself. The body angular momentum $\mathbf{M}$ evolves on a sphere. For three distinct principal moments of inertia, the rotations about the axes of least and greatest inertia are stable on each nonzero sphere, and the middle-axis rotations are unstable.*

By the end of this lecture you will be able to:

- Compute Lie–Poisson brackets and their Casimirs.
- Describe the canonical charts of a nonzero angular-momentum sphere.
- Classify steady rotations for three distinct principal inertias by stability on that sphere.

11.1 Poisson structures on $\mathbb{R}^m$

Since Lecture 5 a Hamiltonian system has lived on an open subset of $\mathbb{R}^{2n}$ with canonical coordinates, $\{q_i, p_j\} = \delta_{ij}$ and all other brackets zero. We now keep the algebra of Lecture 5 — bilinear, antisymmetric, Leibniz, Jacobi — and allow coordinates that are not canonical.

Definition 11.1 (Poisson structure)**.** Let $U \subseteq \mathbb{R}^m$ be open. A *Poisson structure* on U is a smooth map $\Pi\colon U \to \mathbb{R}^{m\times m}$ with $\Pi(x)^{\mathsf{T}} = -\Pi(x)$ for every x, such that the bracket

$$\{f, g\}(x) = \nabla f(x)^{\mathsf{T}}\,\Pi(x)\,\nabla g(x) = \sum_{a,b=1}^{m} \Pi_{ab}(x)\,\frac{\partial f}{\partial x_a}\frac{\partial g}{\partial x_b} \tag{11.1}$$

satisfies the Jacobi identity on $C^\infty(U)$. We call Π the *structure matrix* and (U, Π) a *Poisson manifold*. The *Hamiltonian field* of f is $X_f = \Pi\nabla f$.

Three things are new. The dimension m need not be even. The matrix $\Pi(x)$ need not be invertible. And Π may vary from point to point. The old bracket is the case $m = 2n$, $\Pi = \mathbb{J}$.

The algebraic rules are unchanged. Putting $f = x_a$ and $g = x_b$ in (11.1) gives

$$\{x_a, x_b\} = \Pi_{ab}, \tag{11.2}$$

so the structure matrix *is* the table of brackets of the coordinates. Bilinearity and antisymmetry are immediate, and so is Leibniz ($\{f, gh\} = \{f, g\}\,h + g\,\{f, h\}$), because (11.1) differentiates g once. Only the Jacobi identity is a condition, and by Remark 5.6 it holds

for all observables as soon as it holds for the coordinate functions. As in Definition 6.1, $X_f h = \nabla h^{\mathsf{T}} \Pi \nabla f = \{h, f\}$, so the motion generated by H is

$$\dot{x} = \Pi(x)\, \nabla H(x), \qquad \dot{f} = \{f, H\}, \tag{11.3}$$

and H is conserved along it, since $\{H, H\} = 0$ by antisymmetry.

An invertible $\Pi(x)$ forces m even, since $\det \Pi = \det \Pi^{\mathsf{T}} = (-1)^m \det \Pi$. This lecture is about the other case, a *singular* $\Pi(x)$.

11.2 The Lie–Poisson bracket, and where it comes from

We begin with the angular-momentum bracket.

Definition 11.2 (The Lie–Poisson bracket)**.** On $\mathbb{R}^3$, with the point written $\mathbf{L} = (L_1, L_2, L_3)$, the *Lie–Poisson bracket* is

$$\{f, g\}(\mathbf{L}) = \mathbf{L} \cdot \big(\nabla f \times \nabla g\big), \qquad \text{that is} \qquad \Pi_{ab}(\mathbf{L}) = \sum_c \varepsilon_{abc} L_c. \tag{11.4}$$

The structure matrix is *linear* in $\mathbf{L}$, and its coordinate brackets are

$$\{L_1, L_2\} = L_3, \qquad \{L_2, L_3\} = L_1, \qquad \{L_3, L_1\} = L_2, \tag{11.5}$$

which is (5.11): the brackets of the three components of angular momentum. From (11.4), $X_f = \Pi \nabla f$ works out to

$$X_f(\mathbf{L}) = \nabla f(\mathbf{L}) \times \mathbf{L}. \tag{11.6}$$

In particular $X_{L_3} = e_3 \times \mathbf{L}$: *the third component generates the rotation about the third axis*, counter-clockwise seen from above and at unit angular speed.

Proposition 11.3 (The Lie–Poisson bracket is a Poisson structure)**.** (11.4) *satisfies the Jacobi identity.*

Proof. By Remark 5.6 it is enough to check it on the coordinates. For constant $u, v \in \mathbb{R}^3$, (11.4) gives $\{u \cdot \mathbf{L}, v \cdot \mathbf{L}\} = (u \times v) \cdot \mathbf{L}$, so the cyclic sum of $\{\{u \cdot \mathbf{L}, v \cdot \mathbf{L}\}, w \cdot \mathbf{L}\}$ is $\big[(u \times v) \times w + (v \times w) \times u + (w \times u) \times v\big] \cdot \mathbf{L} = 0$, by the Jacobi identity of the cross product. Take $u, v, w = e_a, e_b, e_c$. ∎

This bracket is what the canonical bracket leaves behind when only the angular momentum is kept. To say so we need maps that carry one bracket to another.

Definition 11.4 (Poisson map)**.** Let (U, Π) and (V, Π') be Poisson manifolds. A smooth $F\colon U \to V$ is a *Poisson map* if $\{f \circ F, g \circ F\}_U = \{f, g\}_V \circ F$ for all $f, g \in C^\infty(V)$.

Theorem 11.5 ($q \times p$ is a Poisson map)**.** *The map $J\colon \mathbb{R}^6 \to \mathbb{R}^3$, $J(q, p) = q \times p$, is a Poisson map from the canonical bracket to* (11.4).

Proof. By the chain rule, for $f \in C^\infty(\mathbb{R}^3)$, $\partial(f \circ J)/\partial q_i = \sum_a (\partial_a f \circ J)\, \partial J_a / \partial q_i$, and likewise in p. Substituting into (5.1) and collecting,

$$\{f \circ J, g \circ J\} = \sum_{a,b} (\partial_a f \circ J)(\partial_b g \circ J)\ \{J_a, J_b\}.$$

By (5.11), $\{J_a, J_b\} = \sum_c \varepsilon_{abc} J_c$, so the right-hand side is $\big[\mathbf{L} \cdot (\nabla f \times \nabla g)\big] \circ J$ — the function of $\mathbf{L}$ in the square bracket, evaluated at $\mathbf{L} = J(q, p)$ — which is $\{f, g\} \circ J$ for the bracket (11.4). ∎

11.3 Casimirs and leaves

Definition 11.6 (Casimir). A function $C \in C^\infty(U)$ is a *Casimir* of a Poisson structure if $\{C, f\} = 0$ for every $f \in C^\infty(U)$, equivalently if $X_C = 0$.

A Casimir is conserved by *every* Hamiltonian, since $\dot{C} = \{C, H\} = 0$ whatever H is. Its Hamiltonian vector field vanishes, so its flow is the identity. In contrast, a nonconstant symmetry charge may generate a nontrivial flow. Proposition 5.9 showed that on a connected canonical phase space $\{C, f\} = 0$ for all f forces C constant, and the same argument works wherever Π is invertible, since $X_C = \Pi \nabla C = 0$ then gives $\nabla C = 0$. Degeneracy is what makes the notion non-trivial.

Proposition 11.7 (The Casimirs of (11.4)). *A function $C \in C^\infty(\mathbb{R}^3)$ is a Casimir of the Lie–Poisson bracket if and only if $\mathbf{L} \times \nabla C(\mathbf{L}) = 0$ for every $\mathbf{L}$, and that holds if and only if C is constant on every sphere $|\mathbf{L}| = R$. In particular $|\mathbf{L}|^2$ is a Casimir.*

Proof. By the triple-product identity, $\{C, f\} = \mathbf{L} \cdot (\nabla C \times \nabla f) = \nabla f \cdot (\mathbf{L} \times \nabla C)$. This vanishes for every f if and only if $\mathbf{L} \times \nabla C = 0$ (take $f = L_a$ for the three values of a). Suppose that holds. At $\mathbf{L} \neq 0$ it says $\nabla C(\mathbf{L})$ is a multiple of $\mathbf{L}$, so the derivative of C in every direction tangent to the sphere through $\mathbf{L}$ is zero; joining two points of that sphere by a smooth path on it and differentiating C along the path shows C is constant on it. The sphere of radius 0 is a point. Conversely, if C is constant on every sphere then its derivative along every tangent direction vanishes, so ∇C is radial and $\mathbf{L} \times \nabla C = 0$. Finally $\nabla |\mathbf{L}|^2 = 2\mathbf{L}$. ∎

So every motion, whatever its Hamiltonian, stays on the sphere $|\mathbf{L}| = R$ on which it starts. That sphere ought to be a phase space with one degree of freedom. But it is not an open subset of $\mathbb{R}^2$, and no single system of coordinates covers it.

Phase space by canonical charts

A chart is a coordinate map on a patch as in Lecture 2, continuous in both directions.

Definition 11.8 (Chart). A *chart* on a set $S \subseteq \mathbb{R}^N$ is a bijection $\kappa \colon U \to V$ from a subset $U \subseteq S$ that is *open in S* (of the form $S \cap O$ with $O \subseteq \mathbb{R}^N$ open) onto an open set $V \subseteq \mathbb{R}^k$, continuous in both directions.

Proposition 11.9 (No chart covers the sphere). *Let $S^2_R = \{\mathbf{L} \in \mathbb{R}^3 : |\mathbf{L}| = R\}$ with $R > 0$. There is no chart $\kappa \colon S^2_R \to V$ with $V \subseteq \mathbb{R}^2$ open.*

Proof. The first component κ_1 of κ is a continuous function on the closed, bounded set $S^2_R \subseteq \mathbb{R}^3$, so it attains a maximum at some point P. But V is open, so it contains $\kappa(P) + (\varepsilon, 0)$ for small $\varepsilon > 0$, and this is κ of a point where κ_1 is larger. ∎

As in Lecture 2, we use overlapping coordinate patches. Here their transition maps must be *canonical*.

Definition 11.10 (Phase space). Let $\mathcal{M} \subseteq \mathbb{R}^N$. A *canonical atlas* of dimension $2n$ on $\mathcal{M}$ is a family of charts $\kappa_a \colon U_a \to V_a \subseteq \mathbb{R}^{2n}$ whose domains cover $\mathcal{M}$, such that for every pair a, b the *transition map* $\tau_{ba} = \kappa_b \circ \kappa_a^{-1} \colon \kappa_a(U_a \cap U_b) \to \kappa_b(U_a \cap U_b)$ is smooth and canonical (Definition 8.1). The coordinates (q, p) of a chart are *canonical coordinates*. A set with a canonical atlas is a *phase space*, or *symplectic manifold*.

Now the sphere. Away from the poles $(0, 0, \pm R)$ use Archimedes' coordinates, the *azimuth* φ and the *height* $z = L_3$:

$$\mathbf{L} = \big(\rho\cos\varphi,\ \rho\sin\varphi,\ z\big), \qquad \rho = \sqrt{R^2 - z^2}, \qquad |z| < R. \tag{11.7}$$

An angle needs two ranges, $(-\pi, \pi)$ and $(0, 2\pi)$, to reach every azimuth, so "the chart (φ, z)" is really two charts differing by the shift $\varphi \mapsto \varphi + 2\pi$. A second pair of charts uses the first axis: the height is L_1, and ψ is the angle about the first axis measured from e_2 towards e_3, so that $\mathbf{L} = (L_1, \rho\cos\psi, \rho\sin\psi)$ with $\rho = \sqrt{R^2 - L_1^2}$. The resulting chart is the first one composed with the cyclic relabelling $(L_1, L_2, L_3) \mapsto (L_2, L_3, L_1)$, and it misses only $(\pm R, 0, 0)$, so between them these charts cover the sphere (Figure 11.1).

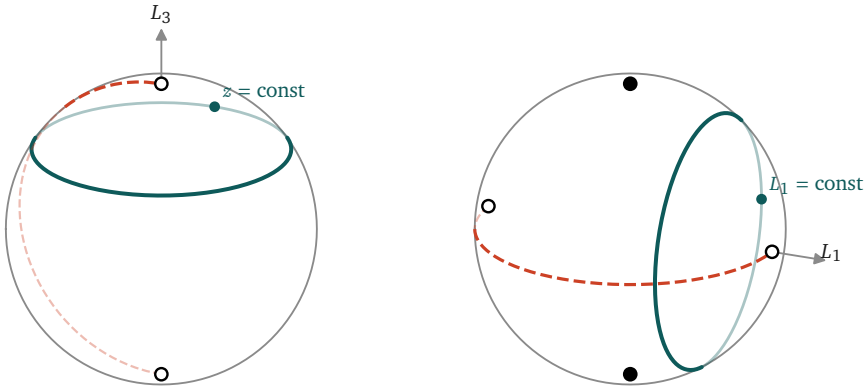


Figure 11.1: Left: the chart (φ, z), with a circle of constant height. It misses the poles (open dots) and the dashed cut. Right: the chart (ψ, L_1), which misses its own poles and cut, so it covers the poles of the first chart (solid dots).

Theorem 11.11 (Canonical coordinates on the Lie–Poisson spheres · *proved; the completeness in (1) uses T1 (cited)*)**.** *Let $R > 0$. For the Lie–Poisson bracket (11.4):*

1. *every Hamiltonian flow maps S^2_R to itself and is complete there;*
2. *for $\mathbf{L} \in S^2_R$ and f, g smooth near $\mathbf{L}$, the value $\{f, g\}(\mathbf{L})$ depends only on the restrictions of f and g to the sphere through $\mathbf{L}$;*
3. *in the chart (φ, z) this bracket is the canonical bracket of $q = \varphi$, $p = z$, so that*

$$\{\varphi, z\} = +1, \tag{11.8}$$

   and in the chart (ψ, L_1) it is the canonical bracket of $q = \psi$, $p = L_1$. These charts are therefore a canonical atlas, and S^2_R is a phase space of dimension 2;
4. *$\Pi(0) = 0$: every X_f vanishes at the origin, which is therefore an orbit by itself.*

So $\mathbb{R}^3$, which is odd-dimensional and has no canonical coordinates, is the disjoint union of the spheres and the origin, and on each piece the Lie–Poisson bracket is the bracket of a phase space. These pieces are the symplectic leaves*, and here the Casimir $|\mathbf{L}|^2$ labels them.*

Proof. (1) $|\mathbf{L}|^2$ is a Casimir (Proposition 11.7), so it is conserved by every flow, and a motion that starts on S^2_R stays on it as long as it exists; S^2_R is compact, so by the blow-up alternative of

Theorem 1.3 the motion exists for all time, and Theorem 1.6 makes the flow a one-parameter group of maps of S^2_R.

(2) $\mathbf{L}\cdot(\nabla f\times\nabla g)$ is unchanged if ∇f or ∇g is altered by a multiple of $\mathbf{L}$. Two smooth functions that agree on the sphere through $\mathbf{L}$ have gradients at $\mathbf{L}$ that differ by such a multiple, since their difference vanishes on the sphere.

(3) The azimuth φ and the height L_3 are also smooth functions on an open subset of $\mathbb{R}^3$, with $\nabla\varphi=(-L_2,L_1,0)/(L_1^2+L_2^2)$, so by (11.4)

$$\{\varphi,L_3\}=\mathbf{L}\cdot\big(\nabla\varphi\times e_3\big)=\mathbf{L}\cdot\frac{(L_1,L_2,0)}{L_1^2+L_2^2}=1.$$

Let f,g have local expressions $F(\varphi,z)$, $G(\varphi,z)$ on the sphere. By (2) we may replace f by $F(\varphi,L_3)$ and g by $G(\varphi,L_3)$, and then the chain rule and antisymmetry give

$$\{f,g\}=\Big(\frac{\partial F}{\partial\varphi}\frac{\partial G}{\partial z}-\frac{\partial F}{\partial z}\frac{\partial G}{\partial\varphi}\Big)\{\varphi,L_3\}=\frac{\partial F}{\partial\varphi}\frac{\partial G}{\partial z}-\frac{\partial F}{\partial z}\frac{\partial G}{\partial\varphi},$$

the canonical bracket of F and G. The same computation with the indices relabelled cyclically, $1\to2\to3\to1$, which leaves ε_{abc} unchanged, gives $\{\psi,L_1\}=1$ and the same conclusion in the second chart.

For the atlas, the two ranges of φ (or of ψ) differ by a translation, which is canonical. The transition τ from (φ,z) to (ψ,L_1) is smooth, and its two components are the local expressions of ψ and L_1 in the first chart. By what was just shown their canonical bracket is $\{\psi,L_1\}=1$. This is the coordinate-bracket condition of Theorem 8.2 for $n=1$, so τ is canonical. The transition the other way is canonical by the same argument with $\{\varphi,L_3\}=1$.

(4) is (11.4) at $\mathbf{L}=0$. ∎

Figure 11.2 draws these leaves. The word applies to any Poisson structure: its *symplectic leaves* are the maximal connected sets swept out by its Hamiltonian flows. We need only the case above.

11.4 The rigid body and its sign

A rigid body with a point fixed at the origin has configuration space $\mathrm{SO}(3)$ (§ 2.2), so its phase space $T^*\mathrm{SO}(3)$ is six-dimensional. Its kinetic energy is unchanged if the whole body is rotated about the fixed point, and for a free body the three components of its spatial angular momentum are conserved (§ 11.5 sketches why). Keeping only the angular momentum leaves three variables and a degenerate bracket on $\mathbb{R}^3$, as $q\times p$ did. First we define those variables.

The body variables

Let the body have a point fixed at the origin and be made of mass points m_a at positions X_a *in the body*, not all on one line through the origin. Its *attitude* at time t is the rotation $R(t)\in\mathrm{SO}(3)$ that carries the reference placement to the current one, so the spatial positions are $x_a(t)=R(t)X_a$. Differentiating $R^{\mathsf T}R=\mathbb{1}$ gives $\dot R^{\mathsf T}R+R^{\mathsf T}\dot R=0$, so $R^{\mathsf T}\dot R$ is antisymmetric, and by Proposition 10.2 there is a unique $\Omega(t)\in\mathbb{R}^3$ — the *body angular velocity* — with

$$R^{\mathsf T}\dot R=\hat\Omega,\qquad \hat\Omega u=\Omega\times u,\qquad \text{hence}\qquad \dot x_a=R\,(\Omega\times X_a).\tag{11.9}$$

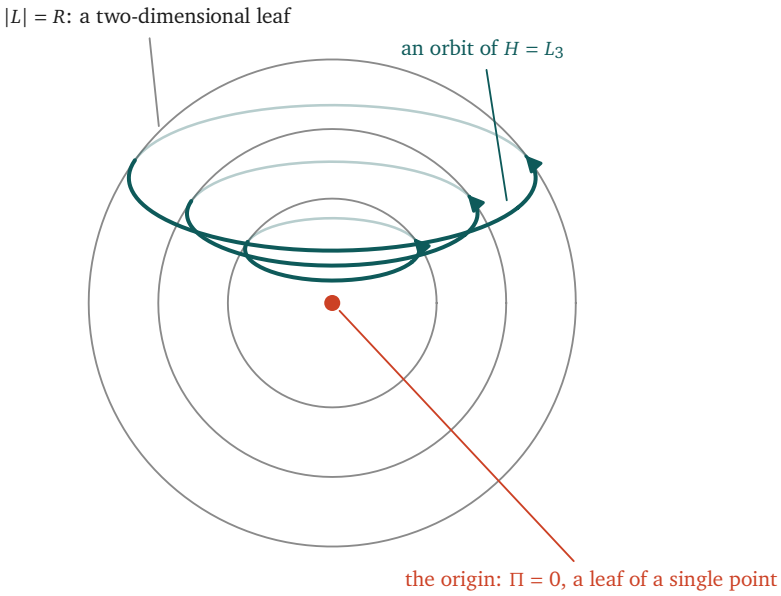


Figure 11.2: The symplectic leaves of (11.4): the spheres $|\mathbf{L}| = R$, each a two-dimensional phase space, and the origin, a leaf consisting of one point, where $\Pi = 0$. On each sphere one orbit of $H = L_3$ is drawn.

Lemma 11.12 (The inertia tensor). *The* inertia tensor *is the linear map* $\mathbb{I}\colon \mathbb{R}^3 \to \mathbb{R}^3$ *given by*

$$\mathbb{I}\Omega = \sum_a m_a X_a \times (\Omega \times X_a).$$

It is symmetric, it is positive definite because the X_a *do not all lie on one line through the origin, and*

$$T = \frac{1}{2}\sum_a m_a |\dot{x}_a|^2 = \frac{1}{2}\,\Omega \cdot \mathbb{I}\Omega, \qquad \mathbf{L} := \sum_a m_a\, x_a \times \dot{x}_a = R\,\mathbb{I}\Omega.$$

Proof. By the triple-product identity $u \cdot \mathbb{I}v = \sum_a m_a (u \times X_a)\cdot(v \times X_a)$: symmetric, and positive at $u = v \neq 0$ unless every X_a lies on $\mathbb{R}u$. Then $|\dot{x}_a| = |\Omega \times X_a|$ gives T, and $Ru \times Rv = R(u \times v)$ gives $\mathbf{L}$. ∎

Write $\mathbf{M} := R^{\mathsf{T}}\mathbf{L} = \mathbb{I}\Omega$ for the *body* angular momentum. It is the same vector as $\mathbf{L}$, read in axes fixed in the body. For a free body $\mathbf{L}$ stays fixed in space, and $\mathbf{M}$ generally changes because the axes turn (Figure 11.3). The exception is a steady rotation about a principal axis, where $\mathbf{M}$ stays constant (Proposition 11.17).

In principal axes — the orthonormal eigenbasis of $\mathbb{I}$ — $\mathbb{I} = \operatorname{diag}(I_1, I_2, I_3)$ with $I_i > 0$, so

$$M_i = I_i \Omega_i, \qquad T = \frac{1}{2}\sum_i I_i \Omega_i^2 = \frac{1}{2}\sum_i \frac{M_i^2}{I_i}. \tag{11.10}$$

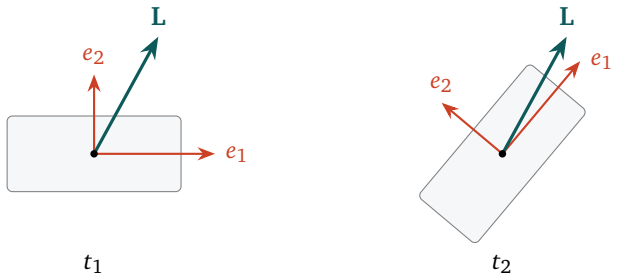


Figure 11.3: A body turning about its fixed point, at two times. The spatial vector $\mathbf{L}$ does not move. The body axes e_1 and e_2 turn with the body, so the components of $\mathbf{L}$ along them, which form $\mathbf{M}$, change.

The reduction

We state the reduction and do not prove it, since nothing in this course builds $T^*\mathrm{SO}(3)$.

Proposition 11.13 (Reduction of the rigid body · *cited*)**.** *Dividing the rotations out of the rigid body's phase space leaves the* body *angular momentum* $\mathbf{M}$ *as its state, with the Poisson structure* $(\mathbb{R}^3, -\Pi)$*, that is,*

$$\{f, g\}(\mathbf{M}) = -\mathbf{M} \cdot (\nabla f \times \nabla g). \tag{11.11}$$

Read in the spatial *angular momentum* $\mathbf{L}$ *instead, the bracket is* (11.4)*, as for* $q \times p$*. (Marsden–Ratiu §§13.1–13.4 and §15.3, pp. 489–491; their conventions are ours.)*

The two readings answer to two different actions of rotations on the body. The vector $\mathbf{M}$ is unchanged when the whole body is rotated in space (R to gR), and $\mathbf{L}$ is unchanged when the body axes are relabelled (R to Rh).

The body bracket carries a minus sign. The map $\mathbf{L} \mapsto -\mathbf{L}$ is a Poisson map from (11.4) to (11.11). We check the sign independently in § 11.5 by deriving the motion of $\mathbf{M}$ from rigid-body kinematics. Only the body bracket reproduces that equation.

Corollary 11.14 (The body leaves)**.** *For the body bracket* (11.11)*,* $X_f = \mathbf{M} \times \nabla f$*, and the Casimirs, the leaves and their canonical charts are those of Proposition 11.7 and Theorem 11.11, carried over by* $\mathbf{L} \mapsto -\mathbf{L}$*, which maps each sphere to itself. In particular* $\{\varphi, M_3\} = -1$*: the canonical chart is* $(\varphi, -M_3)$*.*

Proof. We have $X_f = -\Pi \nabla f = -\nabla f \times \mathbf{M}$ by (11.6). A Poisson bijection carries Casimirs to Casimirs, flows to flows and canonical charts to canonical charts, and $-\mathbf{L}$ lies on the sphere of $\mathbf{L}$; and $\varphi(-\mathbf{L}) = \varphi(\mathbf{L}) \pm \pi$, $L_3(-\mathbf{L}) = -L_3(\mathbf{L})$. ∎

11.5 Euler's equations

Proposition 11.15 (Euler's equation, from kinematics)**.** *Let* $R(t) \in \mathrm{SO}(3)$ *be a smooth motion whose spatial angular momentum* $\mathbf{L} = R\,\mathbb{I}\Omega$ *is constant. Then* $\mathbf{M} = R^{\mathsf{T}}\mathbf{L}$ *satisfies*

$$\dot{\mathbf{M}} = \mathbf{M} \times \Omega, \qquad \Omega_i = M_i / I_i. \tag{11.12}$$

Proof. We have $\dot{\mathbf{M}} = \dot{R}^{\mathsf{T}}\mathbf{L} = (R\hat{\Omega})^{\mathsf{T}}\mathbf{L} = \hat{\Omega}^{\mathsf{T}}R^{\mathsf{T}}\mathbf{L} = -\hat{\Omega}\mathbf{M} = -\Omega \times \mathbf{M} = \mathbf{M} \times \Omega$, using (11.9) and the antisymmetry of $\hat{\Omega}$. ∎

The hypothesis is the only physical input, and we sketch why it holds for a free body. Let no external force act except the one that holds the fixed point, which has no torque about that point. The internal forces of a rigid body cancel in pairs along the lines joining the points — Newton's third law in its strong form — so the total torque about the fixed point is zero and $\mathbf{L}$ is constant.

Euler's equations are Lie–Poisson

Theorem 11.16 (Euler's equations). *On $\mathbb{R}^3$ with the body bracket (11.11) and*

$$H(\mathbf{M}) = \frac{1}{2}\Big(\frac{M_1^2}{I_1} + \frac{M_2^2}{I_2} + \frac{M_3^2}{I_3}\Big),$$

the evolution $\dot f = \{f, H\}$ is exactly (11.12), that is

$$\dot M_1 = \frac{I_2 - I_3}{I_2 I_3} M_2 M_3, \qquad \dot M_2 = \frac{I_3 - I_1}{I_3 I_1} M_3 M_1, \qquad \dot M_3 = \frac{I_1 - I_2}{I_1 I_2} M_1 M_2. \tag{11.13}$$

The spatial bracket (11.4) and the same H would give $\dot{\mathbf{M}} = \Omega \times \mathbf{M}$, the time reverse.

Proof. By Corollary 11.14 the Hamiltonian field is $\mathbf{M} \times \nabla H$, and $\nabla H = \Omega$ by (11.10). Writing out $\mathbf{M} \times \Omega$ componentwise gives (11.13). With Π in place of $-\Pi$ the field is $\nabla H \times \mathbf{M} = \Omega \times \mathbf{M}$ by (11.6). ∎

So the free rigid body is a Hamiltonian system on $\mathbb{R}^3$, with a degenerate bracket and three equations in place of six. It has two conserved quantities of different kinds: the Casimir $|\mathbf{M}|^2$, conserved because of the bracket, and H, conserved because it is the Hamiltonian. The motion runs along the intersection of a sphere with an energy ellipsoid.

Proposition 11.17 (The six equilibria). *Let $I_1 < I_2 < I_3$ and $R > 0$. The equilibria of (11.12) on the leaf $|\mathbf{M}| = R$ are exactly the six points $\pm Re_1, \pm Re_2, \pm Re_3$: the rotations about the principal axes.*

Proof. The a-th component of $\mathbf{M} \times \Omega$ is $\big(1/I_c - 1/I_b\big) M_b M_c$ with (a, b, c) cyclic. The three moments are distinct, so all three components vanish if and only if $M_b M_c = 0$ for each of the three pairs, that is, if and only if at most one M_i is non-zero. On $|\mathbf{M}| = R$ that leaves the six listed points. ∎

11.6 The stable axes

Each leaf is a two-dimensional phase space, so on a leaf the rigid body has one degree of freedom, and the question of Lecture 1 can be asked: which equilibria are stable? Throughout this section and the next $I_1 < I_2 < I_3$ and $R > 0$, and "stable on the leaf" is Definition 1.14 with neighbourhoods taken in the sphere S_R^2.

Theorem 11.18 (The axes of least and greatest inertia are stable on the leaf · *proved; the completeness uses T1 (cited)*). *On S_R^2 the function H attains its maximum $R^2/(2I_1)$ exactly at $\pm Re_1$ and its minimum $R^2/(2I_3)$ exactly at $\pm Re_3$. Each of these four points is a stable equilibrium on the leaf.*

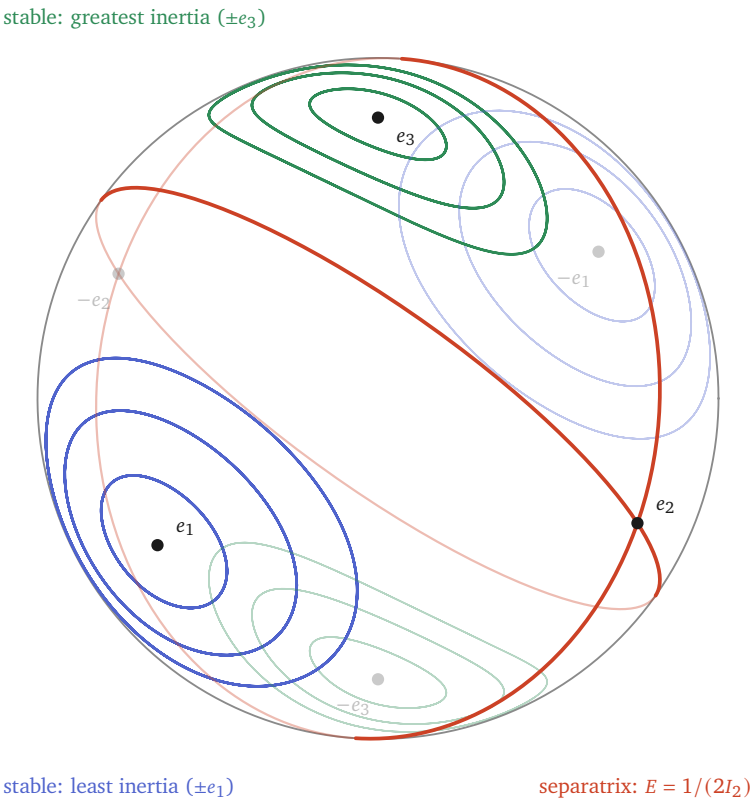


Figure 11.4: The free rigid body on the leaf $|\mathbf{M}| = 1$, with $I = (1, 2, 3)$. Every blue and green curve is an orbit of (11.12). Blue: the orbits round $\pm e_1$, the axis of *least* inertia, where H is largest. Green: the orbits round $\pm e_3$, of *greatest* inertia, where H is smallest. Red: the level $H = 1/(2I_2)$, two great circles drawn from their own equation $M_3 = \pm kM_1$ (§ 11.7), both passing through the two equilibria $\pm e_2$. The far hemisphere is faint.

Proof. On $|\mathbf{M}| = R$,

$$2H - \frac{R^2}{I_1} = \sum_i M_i^2\Big(\frac{1}{I_i} - \frac{1}{I_1}\Big) = M_2^2\Big(\frac{1}{I_2} - \frac{1}{I_1}\Big) + M_3^2\Big(\frac{1}{I_3} - \frac{1}{I_1}\Big) \le 0,$$

with equality only if $M_2 = M_3 = 0$, that is at $\pm Re_1$; and symmetrically $2H - R^2/I_3 = M_1^2(1/I_1 - 1/I_3) + M_2^2(1/I_2 - 1/I_3) \ge 0$ with equality only at $\pm Re_3$. The four points are equilibria by Proposition 11.17.

Stability is Dirichlet's argument: a conserved function with a strict extremum traps the motion. Take $x_* = Re_1$ and $\varepsilon \in (0, 2R)$; a smaller δ serves a larger ε, so this is no restriction. The set $K = \{\mathbf{M} \in S_R^2 : |\mathbf{M} - x_*| = \varepsilon\}$ is closed and bounded and contains neither Re_1 nor $-Re_1$, so $H < R^2/(2I_1)$ on K and its maximum m there is still less than $R^2/(2I_1)$. By continuity choose $\delta \in (0, \varepsilon)$ with $H > m$ at every point of the sphere within δ of x_*. A motion starting there remains on the sphere for all time by Theorem 11.11 and Corollary 11.14. Since H is conserved, $H > m$ for all t and the motion never meets K. Its distance from x_* is continuous in t and starts below ε, so it stays below ε for all $t \ge 0$. The same argument applies at $-Re_1$ and, with reversed inequalities, at $\pm Re_3$. ∎

11.7 The unstable axis

At the middle axis H has neither a maximum nor a minimum on the sphere. On $|\mathbf{M}| = R$,

$$2H - \frac{R^2}{I_2} = M_1^2\Big(\frac{1}{I_1} - \frac{1}{I_2}\Big) - M_3^2\Big(\frac{1}{I_2} - \frac{1}{I_3}\Big),$$

which takes both signs arbitrarily near $\pm Re_2$ and vanishes exactly on the two great circles $\Gamma_\pm = \{|\mathbf{M}| = R,\ M_3 = \pm kM_1\}$, with $k = \sqrt{(1/I_1 - 1/I_2)/(1/I_2 - 1/I_3)} > 0$, both through $\pm Re_2$ (Figure 11.4).

Corollary 11.19 (The middle axis is unstable · *proved modulo T1 (cited)*)**.** *Neither $+Re_2$ nor $-Re_2$ is a stable equilibrium on the leaf.*

Proof. We follow a motion along Γ_- and show that it leaves a fixed neighbourhood of Re_2 in finite time, however close to Re_2 it starts.

First, the motion stays on Γ_-. A motion that starts on Γ_- away from $\pm Re_2$ keeps the value of H, so it stays on $\Gamma_+ \cup \Gamma_-$. These circles meet only at $\pm Re_2$. Those points are equilibria, so by uniqueness (Theorem 1.3) the motion cannot reach them.

Next, an equation for M_2. On Γ_- we have $M_3 = -kM_1$, and the sphere equation gives $(1+k^2)M_1^2 = R^2 - M_2^2$. So (11.13) gives

$$\dot{M}_2 = \Big(\frac{1}{I_1} - \frac{1}{I_3}\Big)M_3M_1 = -c\,\big(R^2 - M_2^2\big), \qquad c = \frac{(1/I_1 - 1/I_3)\,k}{1+k^2} > 0.$$

Finally, the escape. Start on Γ_- with $R/2 < M_2(0) < R$. Then M_2 decreases. While $M_2 \geq R/2$ we have $M_2 \leq M_2(0)$, so $\dot{M}_2 \leq -c\big(R^2 - M_2(0)^2\big) < 0$, a fixed negative rate. So M_2 reaches $R/2$ in finite time, and then the motion is at distance at least $R/2$ from Re_2. Such starting points exist arbitrarily close to Re_2, so Re_2 is not stable. On Γ_+ the same equation holds with the opposite sign, and the same argument applies at $-Re_2$. ∎

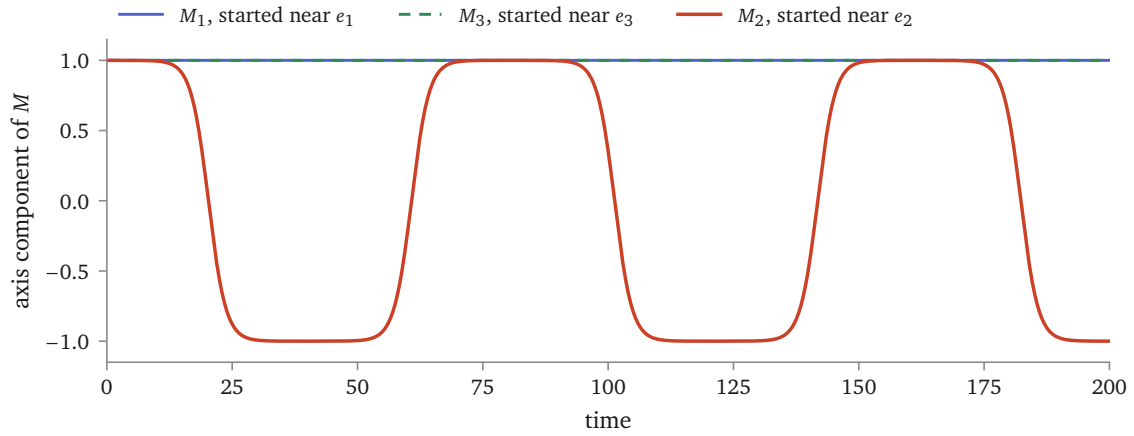


Figure 11.5: Here $I = (1, 2, 3)$ and $|\mathbf{M}| = 1$; each run started close to a principal axis. The axis component of $\mathbf{M}$ stays close to 1 for the two extreme axes (the e_3 run dashed, since the two curves nearly coincide); about the middle axis it changes sign five times in 200 time units.

Near Re_2 but off $\Gamma_\pm$ the motion runs close beside them, from near one middle-axis equilibrium to near the other and back (Figure 11.5). In space $\mathbf{L}$ stays fixed and the body turns

over, repeatedly. This is the *tennis-racket* effect. A racket or a book thrown spinning nearly about its middle axis turns over in the air if the flight lasts long enough; about the other two axes it does not.

Summary.

- *Handed on:* Poisson structures (Definition 11.1); the Lie–Poisson bracket and its Casimirs (Proposition 11.7); its leaves, each sphere a phase space covered by canonical charts and by no single chart (Definition 11.10, Theorem 11.11, and Proposition 11.9); the free rigid body, $\dot{\mathbf{M}} = \mathbf{M} \times \Omega$ for the body bracket (Theorem 11.16). For three distinct principal moments, on each sphere with $R > 0$, it is stable about its extreme axes and not about the middle one (Theorem 11.18 and Corollary 11.19).
- *Next:* Lecture 12 asks when enough conserved quantities make a motion integrable, and gives the third answer to the course's question.

References

Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, whose conventions are ours: §1.2 (pp. 7–9) for the rigid-body bracket and Euler's equations; Chapter 10 (p. 327 on) for Poisson manifolds, with Casimir functions on p. 335; §15.3 (pp. 489–491) for the free rigid body and the flow on the angular-momentum spheres (Figure 15.3.1). Arnold, *Mathematical Methods of Classical Mechanics*, Appendix 14, p. 456: here "the leaves are concentric spheres and their center at the origin". Tong, *Lectures on Classical Dynamics*, §3.4.3 (pp. 57–58), for the tennis racket.